Steps 1 and 2 compared with the Phase A document

What was changed, what was added, and what still has to be changed. Project 26-2-R-2 - Niran Czuckermann and Shahar Bar. Only step 1 (cleaning) and step 2 (breaking the problem into parts) are covered here.

1. In short

- **Step 1 follows Phase A closely.** Phase A promised four cleaning actions (duplicates, self-loops, temporal consistency, isolated nodes). All four were done. Two of them found nothing to remove.
- **Two real differences in step 1:** the dataset is PubMed-Diabetes instead of Cora, and the temporal check could not be done by year, so it was done structurally.
- **Step 2 covers all 17 functional requirements** of Phase A with 7 blocks. Phase A leaves several formulas open; we fixed them in writing. Two of those fixes differ from what Phase A says (closeness direction, and two forests instead of one).
- **One Phase A requirement has no code yet:** requirement 2, the adjacency matrix.
- Section 4 lists everything that has to be changed or decided.

2. Step 1 - Cleaning the database

Phase A, chapter "Anticipated Challenges - Dataset Noise": "the data cleaning pipeline will include removal of duplicate edges, elimination of self-loops, validation of temporal consistency, and filtering of isolated nodes." The unit-test plan repeats the same list.

Topic	Phase A says	What we did	Change?
Dataset	Cora, "under preliminary review and subject to refinement"	PubMed-Diabetes: 19,717 papers, 44,338 citations (the dataset given for the project). Used as is, no sampling	**Yes.** Phase A text is outdated
Duplicate edges	Remove	Removed. **0 found**	No
Self-loops	Remove	Removed. **3 found** (papers 10751750, 9221759, 1586783)	No
Temporal consistency	Validate (implies checking publication years)	The dataset has **no dates**. Checked structurally: citations must not form loops. **11 pairs of papers cite each other**, which is impossible in time. For each pair we kept the edge from the higher PubMed ID to the lower one (higher ID = newer paper, an assumption). Result: 0 cycles, the graph is a DAG	**Yes.** Method differs, plus one assumption
Isolated nodes	Filter	Filtered. **0 found**	No
Edges to unknown papers	Not mentioned	Added as a safety check. 0 found	Addition
Word-feature columns	Not mentioned	The 500 TF-IDF columns are dropped; the class label is kept but not used; the model uses graph structure only	Addition
Edge direction	"Directed edges represent citations" (direction not spelled out)	A cites B is stored as edge A -> B	Clarification

Final graph: 19,717 papers, 44,324 citations (14 fewer than the raw file), one connected piece, no loops.

What Phase A did not expect

- **80 % of papers cite nothing** (15,840 of 19,717) and 2,047 are never cited. The dataset contains only citations that point to papers inside it. This matters later for closeness and for PageRank (see step 2).
- **The ID rule is only a proxy.** We checked it afterwards: 89.6 % of all citations go from a higher ID to a lower one, so the rule is right most of the time, not always. It affects 11 edges out of 44,338.
- Phase A also mentions "false positive" and "false negative" citations. We cannot detect either in this dataset; they are only simulated later by adding and removing edges.

3. Step 2 - Breaking the problem into parts

Phase A has 17 functional requirements, 12 non-functional requirements and a flow chart in Appendix A. We split the model into 7 blocks (A features, B scaler, C model, D ranking, E perturbation, F stability, G core). Each block is one module in src/ with a clear input and output. Every functional requirement is covered by a block, except requirement 2.

Where Phase A is silent, and the rule we wrote

Open point	Phase A	Our rule	Differs?
Closeness on a directed graph that is not connected	Text: average distance **from a node to all others** (outgoing). Test plan: return 0 when no path exists	Wasserman-Faust formula with **incoming** distances (how close the rest of the network is to the paper). Outgoing gives 0 to 80 % of papers; incoming to 10 %	**Yes** - needs approval
k-core on a directed graph	Only the undirected definition is given	Computed on the undirected version, self-loops removed	Clarification
Betweenness on 19.7k nodes	"NetworkX (or igraph)", no approximation mentioned	Exact, with igraph. We first planned a sampled version, measured it (Spearman 0.55 with the exact one) and dropped it	No (matches Phase A)
Degree	Appendix B: divided by N-1	Total degree / (N-1)	No
Scaler	Min-Max fitted on the original graph, reused on perturbed graphs (requirement 12)	Same. Values may leave [0,1] on perturbed graphs and are not clipped	No
Zero mean in CV	Set instability to 0	Same	No
Which model(s)	One forest; PageRank is both an input feature and the target	**Two forests**: with and without PageRank as an input; the one without is treated as the main model. With PageRank as an input the forest simply copies it	**Yes** - needs approval
Edge addition	Random edges, any two papers	Two versions: the Phase A one, and one that adds only newer -> older edges (no impossible loops)	Addition - needs approval
Training data	Trained on the original graph	Same, all nodes. An 80/20 hold-out check is added as a sanity test	Addition
Random seeds	Fixed seeds (non-functional req. 2)	One project seed (42) plus the run number	No

4. What has to be changed or decided

A. In the text (Phase B report; the Phase A file itself stays as submitted)

#	Change	Where in Phase A
1	Replace Cora with PubMed-Diabetes: size, source, classes, and the dataset reference (Namata et al. 2012 - please verify the reference)	"Identifying a Suitable Dataset"
2	Describe the cleaning with the real numbers (0 duplicates, 3 self-loops, 11 mutual pairs, 0 isolated nodes)	"Dataset Noise and Programmatic Verification"
3	Replace "validation of temporal consistency" with the structural check and the PubMed-ID assumption, including the 89.6 % result	same section; "Scientific Citation Networks Context"
4	Add the second forest (without PageRank) to the feature vector definition, to the training protocol and to the flow chart	"Model Definition" (node representation, training protocol); Appendix A
5	Add the "newer -> older only" edge addition next to random addition	"Graph Perturbation Model"; "Evaluation Plan"
6	State the closeness direction explicitly, and add how unreachable pairs are handled (Wasserman-Faust)	"Global Centrality Measures"; Appendix B (closeness)
7	Say that betweenness is exact (igraph) and why sampling was rejected; list igraph as used, not optional	Development Environment
8	State that k-core is computed on the undirected graph	Appendix B (k-core)
9	Development environment: we worked locally, notebooks also run on Colab; the plan said Colab first	Development Environment (minor)
10	Check the cross-references: requirement 15 points to "Section 3.4", the test plan to "Section 5.1, Requirement 12/17", the environment text to "section 6.2". They may not match the final numbering	Requirements; Testing Plan

B. In the project (code, tests or decisions)

#	Change	Effort
11	**Requirement 2, adjacency matrix:** no code builds one (the graphs are NetworkX and igraph objects). Either add a small function that returns a sparse adjacency matrix, with a test, or ask the advisor to drop the wording. Adding it is cheap and does not touch any result	about 15 minutes
12	**Closeness direction - team decision.** Keep incoming (our recommendation; fix the Phase A text), or switch to outgoing (one flag in features.py, but then everything after step 2 must be rerun)	decision; rerun about 15 minutes if switched
13	**Mutual citations (11 edges):** optional check - rerun once with both edges of each pair removed, to show the PubMed-ID rule does not matter. Not needed if the advisor accepts the assumption	about 10 minutes
14	Non-functional requirement 4 asks for a "predefined size N and M" the system supports. Phase A gives no numbers; we should write down the size we tested (19,717 nodes, 44,324 edges)	one sentence in the report

C. Needs the advisor's agreement

- Dataset change (Cora to PubMed-Diabetes).
- Structural temporal check and the "higher PubMed ID = newer" assumption.
- Closeness direction (item 12).
- Two forests, with the one without PageRank as the main model.
- The extra "newer -> older" edge addition.

Everything above is also recorded in docs/decisions.md (D1, D2, D3, D5, D10, D11).